Unit: 6 Unit 9: Data Statistical Analysis
Topic: Two-Way Tables

Name: ________________
Date: ___________

Advanced (Habanero)

Q1. What is the purpose of a two-way table?
(A) To display one variable
(B) To compare two variables
(C) To calculate mean
(D) To display a single data point
(E) To create a graph

Q2. In a two-way table, what does a cell represent?
(A) A single data point
(B) A group of data points
(C) The mean of the data
(D) The median of the data
(E) The intersection of two variables

Q3. If a two-way table has 2 rows and 3 columns, how many cells are in the table?
(A) 2
(B) 3
(C) 5
(D) 6
(E) 9

Q4. What type of frequency is calculated by dividing the frequency of a cell by the total frequency?
(A) Absolute frequency
(B) Relative frequency
(C) Joint frequency
(D) Marginal frequency
(E) Conditional frequency

Q5. If $P(A) = 0.3$ and $P(B) = 0.4$, what is $P(AcapB)$ if A and B are independent?
(A) 0.12
(B) 0.15
(C) 0.20
(D) 0.25
(E) 0.30

Q6. In a two-way table, what is the purpose of calculating relative frequencies?
(A) To compare the size of the groups
(B) To analyze the association between variables
(C) To create a bar graph
(D) To calculate the mean
(E) To compare the medians

Q7. If a two-way table shows a strong association between two variables, what can be concluded?
(A) The variables are independent
(B) The variables are dependent
(C) The variables are mutually exclusive
(D) The variables are complementary
(E) The variables are inversely proportional

Q8. What is the term for the frequency of a row or column in a two-way table?
(A) Joint frequency
(B) Marginal frequency
(C) Conditional frequency
(D) Relative frequency
(E) Absolute frequency

Open Ended Questions (FRQ)

Q9. Create a two-way table to analyze the relationship between favorite sports (football and basketball) and gender (male and female) for a group of 20 students. Assume 8 males like football, 6 males like basketball, 4 females like football, and 2 females like basketball.

Q10. Using the table from question 9, calculate the relative frequencies and analyze the association between favorite sports and gender.

Chef's Tips

- Remind students to use the given formulas and concepts
- Encourage students to show their work and explain their reasoning
- Emphasize the importance of accuracy and attention to detail